

Problem Set 3: Detailed Answer Key & Derivations

CBE 30235: Introduction to Nuclear Engineering

Problem 1: Carbon Dating Limits

(a) **The "Modern" Limit (1 gram)** The limit is defined where the signal S drops to 2σ (Signal-to-Noise Ratio = 2). First, calculate the standard deviation of the background measurement:

$$\sigma \approx \sqrt{\frac{2B}{t}} = \sqrt{\frac{2(1.0 \text{ cpm})}{1000 \text{ min}}} = \sqrt{0.002} \approx 0.0447 \text{ cpm}$$

The minimum detectable activity (S_{min}) is:

$$S_{min} = 2\sigma = 0.0894 \text{ cpm}$$

Now, use the radioactive decay law $S = S_0 e^{-\lambda t}$ to find the age:

$$\lambda = \frac{\ln 2}{5730} \approx 1.21 \times 10^{-4} \text{ yr}^{-1}$$

$$0.0894 = 13.5 e^{-\lambda t} \implies e^{-\lambda t} = \frac{0.0894}{13.5} \approx 0.00662$$

$$t = -\frac{1}{1.21 \times 10^{-4}} \ln(0.00662) \approx \mathbf{41,470 \text{ years}}$$

(b) Scaling to 10 grams

- New Signal $S'_0 = 10 \times 13.5 = 135 \text{ cpm}$.
- New Background $B' = 1.0 \times (10)^{0.7} \approx 5.01 \text{ cpm}$.

$$\sigma' = \sqrt{\frac{2(5.01)}{1000}} \approx 0.100 \text{ cpm}$$

$$S'_{min} = 2(0.100) = 0.200 \text{ cpm}$$

$$t = -\frac{1}{1.21 \times 10^{-4}} \ln\left(\frac{0.200}{135}\right) = -\frac{1}{1.21 \times 10^{-4}} \ln(0.00148)$$
$$t \approx \mathbf{53,840 \text{ years}}$$

Conclusion: Increasing the sample mass by 10x only extends the clock by about 12,000 years (approx 2 half-lives).

(c) **Diminishing Returns** To extend the age limit by exactly **one half-life**, the minimum detectable signal S_{min} must be reduced by a factor of 2. Since $S_{min} \propto \sigma \propto \frac{1}{\sqrt{t}}$, we have:

$$\frac{1}{2} = \frac{1}{\sqrt{x}} \implies \sqrt{x} = 2 \implies x = 4$$

You must increase the counting time by a factor of **4** (to 4000 minutes) to gain just one half-life of range.

Problem 2: The N/Z Zig-Zag

(a) Alpha Decay Effect Parent: ${}_{92}^{238}\text{U}$ ($N = 146, Z = 92$). $N/Z = 1.587$. Reaction: ${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He}$. Daughter: ${}_{90}^{234}\text{Th}$ ($N = 144, Z = 90$).

$$\text{Ratio} = \frac{144}{90} = \mathbf{1.600}$$

Mathematically, since the starting ratio ($146/92 \approx 1.59$) is less than the ratio of the removed particle ($2/2 = 1.0$), removing the alpha particle leaves the remainder *more* neutron-rich.

(b) Correcting Decay The nucleus is now too neutron-rich. To fix this, it must convert a neutron into a proton.

$$n \rightarrow p + e^- + \bar{\nu}_e$$

The specific mode is **Beta Minus (β^-) decay**.

(c) Total Chain Length Total change in Mass Number A : $238 - 206 = 32$. Since only alphas change A (by 4 units each):

$$N_\alpha = \frac{32}{4} = \mathbf{8} \text{ alpha decays}$$

Total change in Proton Number Z :

$$Z_{\text{final}} = Z_{\text{initial}} - 2(N_\alpha) + 1(N_\beta)$$

$$82 = 92 - 2(8) + N_\beta$$

$$82 = 76 + N_\beta \implies N_\beta = \mathbf{6} \text{ beta decays}$$

Problem 3: Radon Transport

Values used: $\kappa = 1.0 \times 10^{-11} \text{ m}^2$, $A = 100 \text{ m}^2$, $\nabla P = 5 \text{ Pa/m}$.

(a) Darcy Velocity We calculate the velocity of gas rising through the soil *towards* the foundation.

$$v = \frac{\kappa}{\mu} \nabla P = \frac{1.0 \times 10^{-11} \text{ m}^2}{1.8 \times 10^{-5} \text{ Pa} \cdot \text{s}} (5 \text{ Pa/m})$$

$$v \approx 2.78 \times 10^{-6} \text{ m/s}$$

(b) Volumetric Flow Rate Since the gravel layer distributes the suction, gas enters from the soil over the entire footprint area A_{slab} .

$$Q_{\text{soil}} = v \times A_{\text{slab}} = (2.78 \times 10^{-6} \text{ m/s})(100 \text{ m}^2)$$

$$Q_{\text{soil}} = 2.78 \times 10^{-4} \text{ m}^3/\text{s}$$

Convert to hr:

$$Q_{\text{soil}} = 2.78 \times 10^{-4} \times 3600 \approx \mathbf{1.0 \text{ m}^3/\text{hr}}$$

(c) Indoor Concentration Steady state mass balance (Source = Sink):

$$\text{Source} = Q_{\text{soil}} C_{\text{soil}}$$

$$\text{Sink} \approx Q_{\text{vent}} C_{\text{indoor}}$$

$$Q_{vent} = 500 \text{ m}^3 \times 0.5 \text{ hr}^{-1} = 250 \text{ m}^3/\text{hr}$$

Solving for C_{indoor} :

$$C_{indoor} = \frac{Q_{soil}}{Q_{vent}} C_{soil} = \frac{1.0}{250} (2000 \text{ pCi/L})$$

$$C_{indoor} = 0.004 \times 2000 = \mathbf{8.0 \text{ pCi/L}}$$

Conclusion: This level is **double** the EPA Action Level of 4.0 pCi/L. *Physics Note: This illustrates why the "Gravel Layer" is critical in Radon modeling. It acts as a collection manifold (the wide part of the funnel), gathering gas from a large area of soil and channeling it into the house.*

Problem 4: U-Pb Concordia

(a) **U-238 Age**

$$D = P(e^{\lambda t} - 1) \implies t = \frac{1}{\lambda} \ln \left(1 + \frac{D}{P} \right)$$

$$t_{238} = \frac{1}{1.551 \times 10^{-10}} \ln(1 + 0.60) = \mathbf{3.03 \times 10^9 \text{ yr}}$$

(b) **U-235 Age**

$$t_{235} = \frac{1}{9.848 \times 10^{-10}} \ln(1 + 42) = \mathbf{3.82 \times 10^9 \text{ yr}}$$

(c) **Interpretation** The ages disagree (3.03 Ga vs 3.82 Ga), so the Zircon is **Discordant**. **Physical Meaning:** The U-Pb clock relies on the assumption of a closed system. Disagreement usually implies **Lead Loss**. Since Lead is not chemically compatible with the Zircon crystal lattice, a thermal event (metamorphism) can drive Lead out of the crystal. This "resets" the clock partially, making the sample appear younger than it actually is. These numbers correspond to a 4.5Gy old sample with a significant lead loss at 1Gy. You would need multiple zircons from the same sample (with different amounts of lead loss) to determine the two ages.

Solution 5: The Earth's Nuclear Furnace

(a) **Total Uranium Mass** We first calculate the power contribution of ^{238}U :

$$P_{total} = 25 \text{ TW} \implies P_{U238} = 0.40 \times 25 \text{ TW} = 10 \text{ TW} = 10^{13} \text{ W}$$

Using the specific power $h = 9.4 \times 10^{-5} \text{ W/kg}$:

$$M_{U238} = \frac{P_{U238}}{h} = \frac{10^{13} \text{ W}}{9.4 \times 10^{-5} \text{ W/kg}} \approx \mathbf{1.06 \times 10^{17} \text{ kg}}$$

(b) **Integrated Energy Release** Integrating the average power over the age of the Earth:

$$E_{radio} = P_{avg} \times \Delta t$$

$$E_{radio} = (40 \times 10^{12} \text{ J/s}) \times (4.5 \times 10^9 \text{ yr}) \times (3.156 \times 10^7 \text{ s/yr})$$

$$E_{radio} \approx 5.68 \times 10^{30} \text{ J}$$

(c) **Comparison and Interpretation** Comparing the two energy sources:

$$\frac{E_{radio}}{E_{acc}} = \frac{5.68 \times 10^{30} \text{ J}}{2.2 \times 10^{32} \text{ J}} \approx \mathbf{0.026} \quad (\sim 2.6\%)$$

Why is radioactivity critical? Even though the total radiogenic energy is only $\sim 2.6\%$ of the accretion energy, it is the dominant factor in the Earth's *current* geology because of **timing**:

- **Accretion Energy (Flash):** This massive energy was released instantly during formation. Most of it created a magma ocean and was radiated into space immediately. It set the initial temperature but could not maintain it.
- **Radiogenic Energy (Sustained Burn):** This energy is released continuously inside the rock. It acts as a "thermal blanket" (or internal furnace) that offsets the cooling of the planet. Without this continuous power, the mantle would have cooled, viscosity would have increased, and plate tectonics would likely have shut down billions of years ago.